Agent-mediated Self-regulated Learning

A Conceptual Model based on Knowledge Management at Agent Level

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***Abstract*—Continuous improvement and self-regulation are often mentioned when a person talks about quality of education or teaching and learning. In 1990s, the concept of self-regulated learning (SRL) came into picture to describe the need for learners to consistently improve their study method, and it was seen cited again in recent years due to the emergence of new technologies that have opened more windows to the learners to improve their studies. Nevertheless, there is a gap in the research on knowing who is performing the SRL and who is not. This has brought into the attention of presenting this paper on knowing what a learner is doing, using software agents to intelligently trace the learners' online learning activities. The purpose of this paper is to conceptualise an idea of having software agents to mediate the tasks of the learners, by performing personal knowledge management (PKM) processes at agent level. The expected outcome is a conceptual model of agents' interaction with their environment, and how to analyse the information or knowledge retrieved from the sources in order to provide a better report to their learners for their benefit of SRL.**

Keywords—self-regulated learning; personal knowledge management; software agent; nodal approach; social media

I. INTRODUCTION

Success in academic relies a lot on learners' effort in study. Different learners have different drive, approach and expectation towards their academic performance. It is realised that learners need to practice self-regulated learning (SRL) in order to sustain and ensure success in academic. Self-regulated learning would keep the learners persistent in their studies and quest for knowledge. However, there is a challenge in knowing or identifying who practice self-regulated learning among the learners, especially in universities where learners are highly independent in their studies.

On the other aspect of learning, the concept of managing personal knowledge comes into picture as researchers witnessed the patterns of managing knowledge over the emerging technologies like Web 2.0 and mobile applications. The concept of personal knowledge management (PKM) has been the recent research interest when the trend shows that learners turn to the new technologies for almost anything that they want to know, enabling them to connect to anyone they want to as well.

Seeing the relationship between these two trends, an idea emerged for having intelligent software agents to facilitate the self-regulated learning over the Internet and other systems provided by the university. The main purpose is to enable the learners with a capability of knowing their status of progress in study for SRL, enlightening them to react or respond to their status. It is known that the effect on the learners could be positive or negative towards continuous improvement of study, but this paper only looks into the mechanism of the intelligent agents' tasks and reporting details.

The purpose of this paper is to conceptualise an idea of having software agents to mediate the tasks of the learners, by performing PKM processes at agent level. In doing so, the question that needs to be answered is: "How could software agents mediate the tasks of reporting the learner's online learning time?"

II. RELATED WORKS

As mentioned in the previous section, there are two aspects of research covered in this paper: self-regulated learning; and agent-mediated personal knowledge management. This section looks into the previous works related to these domains.

A. Self-regulated Learning

Most research on self-regulated learning (SRL) may have started around 1990, when the self-regulated learners are described as follows:

> *"Self-regulated learners are aware when they know a fact or possess a skill and when they do not. Unlike their passive classmates, self-regulated students proactively seek out information when needed and take the necessary steps to master it. When they encounter obstacles such as poor study conditions, confusing teachers, or abstruse test books, they find a way to succeed."* [1]

By comparing the two sets of learners, i.e. self-regulated learners and passive learners, it is emphasised that self-regulation skill is crucial for learners to take advance and learn well in online learning [2]. Self-regulated learning is defined as "a student's ability to independently and proactively engage in self-motivating and behavioural processes that increase goal attainment" [3]. In other words, self-regulated learners are students who are metacognitively, motivationally and behaviourally active participants in their own learning process [4].

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"Each learning state consists of three cyclical phases that influence each other while exerting an impact on the subsequent learning state." [5] Schmitz and Wiese [6] suggested a process model of self-regulated learning, as shown in Figure 1, in which the three cyclical phases are defined according to labels defined by Heckhausen and Kuhl [7]: pre-action phase; action phase; and post-action phase. Schmitz, Klug and Schmidt [8] described these phases as follows:

- **Pre-action phase** precedes learning. The situation and the assigned task are the source from which the students set goals, develop various attitudes towards learning, such as intrinsic and extrinsic motivations to learn, and develop self-efficacy for managing tasks. The output of this phase becomes the predictor of the learning processes in the next phase.
- During the **action phase**, the quantity and quality of learners' performance matter. Learning strategies focuses on the metacognitive and resource management strategies, such as regulation, effort, time and attention management, or learning with peers. The outcome of this phase is the result observed in the next phase.
- **Post-action phase** focuses on the students' metacognitive and affective reactions to the quantity and quality of their learning outcomes. These outcomes refer to that special learning session at one point of time, and they will influence the next learning session, i.e. back to pre-action phase of the new learning session.

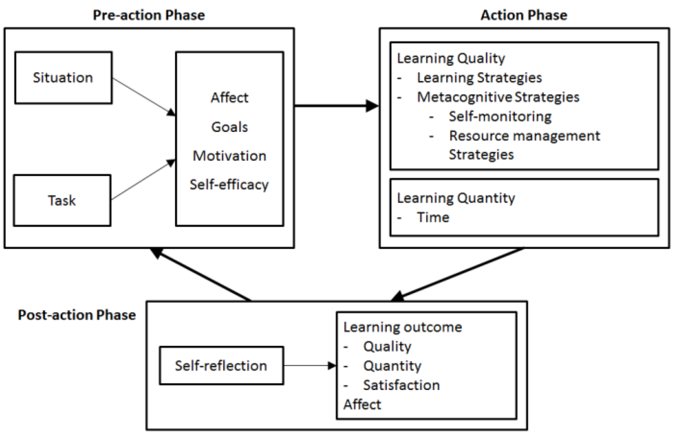

Fig. 1. Process model of self-regulated learning [6].

In most cases, self-regulation skills are correlated to academic performance [1], but it is not formally taught to learners despite its importance, unlike domain knowledge skills like mathematics or arts. Supported by a survey that has proven the learners' way of study is not based on formal teaching [9], it is important to design online learning environment that support self-regulated learning strategies.

Dabbagh and Kitsantas [10] suggested a framework for using social media to support self-regulated learning among students, by defining three levels: Level 1 on personal information management; Level 2 on social interaction and collaboration; and Level 3 on information aggregation and management. The authors have identified how these levels of usage can be implemented on blogs, wikis, Google^{TM} Calendar, YouTube^{TM}, Flickr^{TM}, social networking sites, and social bookmarking.

B. Agent-mediated Knowledge Management

In recent research on agent-mediated personal knowledge management (PKM), the above mentioned social media tools for SRL [10] are the platforms where agents can reside in to mediate the tasks on behalf of their human counterparts [11], in which human could be any knowledge worker, like researcher and learner. As shown in Figure 2, among the social media tools mentioned by Ismail and Ahmad [11] and supported by Dabbagh and Kitsantas [10] include blogs, wikis, social networking sites (Figure 2 listed the functions of this social media tool instead of 'social networking sites', e.g. 'follow', comment, tag, chat and vote), and social bookmarking. Nevertheless, the social media tools such as online calendar (i.e. Google^{TM} Calendar) and multimedia sharing sites (i.e. YouTube^{TM}, Flickr^{TM}) are still relevant to be the platforms for knowledge management processes, with similar features found to be embedded in the earlier mentioned social media tools.

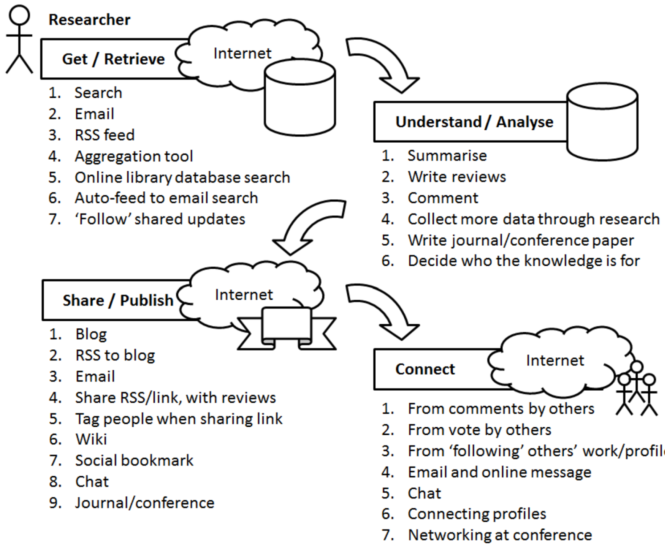

Fig. 2. Overview of personal knowledge management [11].

According to the PKM framework, the mediation of tasks to the software agents include get/retrieve knowledge, understand/analyse knowledge, share knowledge, and connect to knowledge source [12], since software agents are capable of:

- engaging in dialogs to negotiate and coordinate [13];
- perceiving its environment through sensors and acting upon that environment through effectors [14];
- carrying out some set of operations on behalf of a user or another program with some degree of independence or autonomy, and in so doing, employing some knowledge or representation of the user's goals or desires [15];
- sensing and acting autonomously in the dynamic environment that they inhabit, and realising a set of goals or tasks for which they are designed [16, 17]; and

- performing flexible action in the an environment in order to meet its design objectives [18].

Software agents are said to be able to roam the Internet to find knowledge sources that indicate a person's online activities. Derived from Figure 2, Figure 3 shows the overview of this concept using nodal approach, in which the agents are assigned to reactively and proactively get/retrieve information from sources like online database, repositories and knowledge base [11, 19].

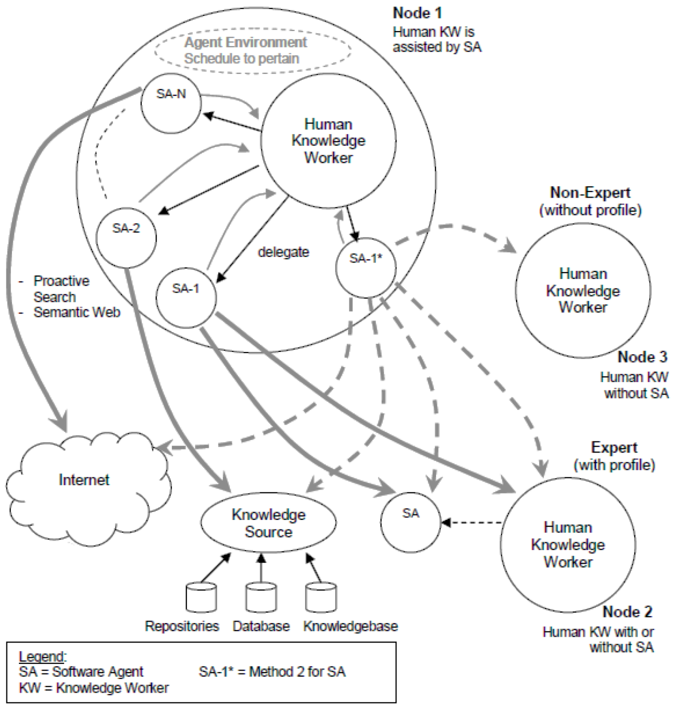

Fig. 3. Multiple nodes of agents mediating personal knowledge management [19].

Based on Figure 3, a node, N, is specified as follows: a knowledge worker, KW; all functions of the knowledge worker, f_i, one or more agents, A_j; and all functions of agents, f_{jk}. In other words, a node is a four-tuple structure:

$$N = < KW, f_i, A_j, f_{jk} > \qquad (1)$$

where $f_i \in KW$, $f_{jk} \in A_j$, and $i, j, k \geq 1$.

In fulfilling the need to coordinate each agent's tasks at granular level, the GUSC Model is suggested by Ismail and Ahmad [20], postulating four main processes in PKM: get/retrieve, understand/analyse, share/publish, and connect to knowledge source. In short, the acronym "GUSC" basically means Get-Understand-Share-Connect. This model has been proven its usability in various research, including those in educational technology [21, 22, 23].

C. Agent-based PKM in SRL

The earlier works of SRL mainly focus on the PKM process model [24, 25] as opposed to the agents of PKM. Schmidt suggested that the social media could facilitate the three phases of SRL (pre-Action, Action and post-action). Turker and Zingel's work was mainly on the first phase whereby the forethought act of organising learning resources into meaningful learning activities towards achieving a set of goals.

A prominent research agent-based PKM in SRL was by Gorman and Pauleen [26] who suggested a holistic perspective of agents in PKM as 'Systems Intelligence' (SI). The theoretical idea was not only SI focuses on the individual elements of a learner, the system also focus on the interconnectedness of these individual elements that ideally caused the individuals excel in a complex system.

Duffy and Azevedo [27] has evaluated the use of agents as a catalyst of motivation within a multi-agent intelligent tutoring system. The research can be associated to the GUSC model where in the 'action' phase, the responsiveness to agent prompts will depend on whether the students feel the strategies will help promote understanding, which is similar to Get/Understand phases in GUSC. The 'share' phase in Duffy and Azevedo [27] is similar to the Share/Connect phases the use of agents will influence the evaluations to convey the information about their competence to others.

III. METHODOLOGY

This research sees the opportunity of designing a multi-agent system to monitor the self-regulated learning at student level, by focusing on the action and post-action phases as shown in Figure 1. It is possible for agents to work together and monitor students' learning time, learning strategies and resource management strategies, from their activities online, and retrieve information on the outcome of the actions from academic results and students' online posts on their satisfaction. In supporting this intention, the framework for using social media to support SRL [10] is referred to.

Few assumptions are made to identify the parameters for the software agents to trace. A set of learners' online activities is defined to relate to the PKM processes that match these parameters. In general, Table I shows the overall view of the processes involved during online learning time and online social time, which are considered as parameters.

From the defined scope in Table I, the nodal approach is used to illustrate the overview of the agents' interactions in their environment in pursuit to get and connect to knowledge sources. This is followed by the calculation for the parameters, for the agents to perform in reporting considerations. The next section presents further details on this.

TABLE I. PKM PROCESSES RELATED TO PARAMETERS

PKM Process	Description	Source / Location	Parameter
Get / Retrieve	Learners would spend time accessing the source to retrieve learning materials, view multimedia files, and download the materials or files	• University learning portal or repositories (i.e. LMS[c], VLE[d], Moodle, etc.) • Social media (i.e. social	• Learning time, T_L • Online social time, T_S

PKM Process	Description	Source / Location	Parameter
Understand / Analyse	Learner would spend time assessing the learning material/file contents on the source to understand or analyse for the benefit of self-learning.	network, blog, microblog, chat, etc.) – could be for learning[a]	• Learning time, T_L • Online social time, T_S
Share	Learners would spend time on the source to share their ideas and knowledge with peers and teacher, be involved in the online discussion forum on topics pertaining the learning contents		• Learning time, T_L • Online social time, T_S
Connect[b]	Learners would spend time on the source, to connect to another person regarding the learning topics, such as add friend, join online discussion group, follow a profile, and tag a person.	• Social media (i.e. social network, blog, microblog, chat, etc.) – could be for learning[a]	Online social time, T_S

a. Requires agent's capability to learn from keywords that reflect certain topics, intent or reflection pertaining learning contents.

b. Source/location for connect is assumed to be for related activities outside the boundary of the university learning systems, which requires the agent to be mobile and able to roam the Internet.

c. LMS = Learning Management System, commonly provided by the university

d. VLE = Virtual Learning Environment, may be provided by the university

IV. Preliminary Findings

In analysing the model in previous work depicted in Figure 3, a more concise model is illustrated for this study. As shown in Figure 4, the software agents may not need to interact with other agents that reside on other human counterparts nor directly with other human counterparts. This is the distinguished difference between this study and the previous work by Ismail and Ahmad [19].

Apart from that, there is a possibility of having multiple routes for a single agent, as shown on agent called SA-N in Figure 4. It would be common to assign single software agent to perform tasks on single route, and by having multiple agents to move to different routes at the same time, to avoid possible conflicts of time and duration of multiple tasks. This is perceived as common since technology-savvy knowledge workers like the new generation of learners is capable of multi-tasking and accessing multiple sites at the same time.

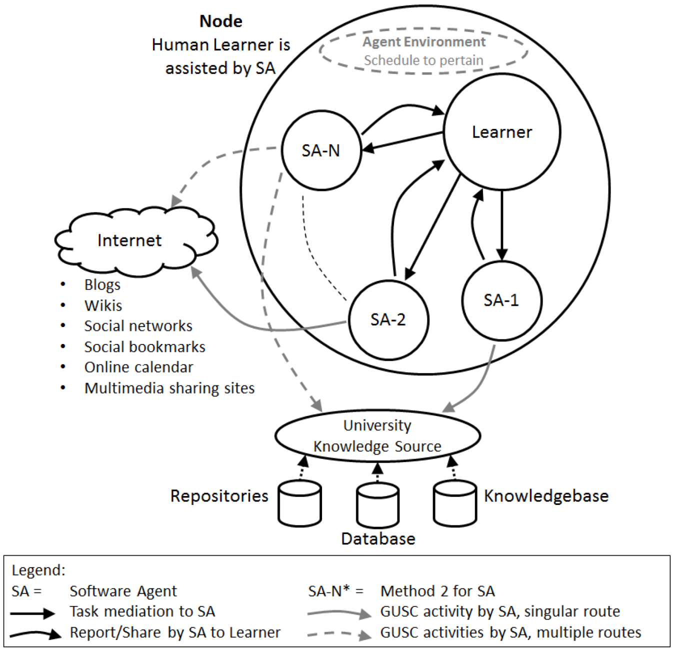

Fig. 4. Multiple nodes of agents mediating GUSC processes for SRL.

Nevertheless, the knowledge source may differ between one case to another, depending on the learners' online behaviour and interactions. This is as mentioned in Figure 1 [6], in which situation is a factor at pre-action phase that affects the individual learner's motivation to perform activities online. Even so, this research is able to zoom in to the activities and potential knowledge sources, in which Table I presented in a very briefly manner. This is as shown in Table II.

Upon enlisting the online learning activities in Table II, it is found that mobile applications (i.e. mobile app) is inevitable to be listed as one of the potential knowledge sources. This is due to the fact that most online tools are made available in the form of mobile app, in order to provide mobility to the users, such as mobile social network, blog app, and mobile version of LMS. Being the generation of this age, the learners are very into mobile devices, in which the trend of accessing knowledge sources anytime anywhere is perceived vital.

Looking at the need for software agents to analyse the right information or knowledge from the sources, the following Table III presents the general idea of how to extract the measurement for each parameter. In this table, idle time (e) and play time (T_P) are introduced, due to the need for the agents to separate the unwanted information from the perceived true learning time.

TABLE III. PKM PROCESSES RELATED TO LEARNING ACTIVITIES

PKM Process	Activities	Potential Knowledge Sources
Get / Retrieve	• Download learning material • View learning material • Follow teacher or other learner • Respond to teacher's or other learner's post • Search for material related to learning topics	• Learning portal, LMS, VLE • Social network • Search engine
Understand / Analyse	• Comment on learning material • Converse or chat with teacher or other learner on learning material • Perform task as per instruction given by the teacher in a short duration after instruction is given • Respond to teacher's or other learner's questions regarding learning topics	• Learning portal, LMS, VLE • Social network • Mobile app
Share	• Upload learning material • Post information regarding learning material • Tag other learner to the shared post or learning material	• Learning portal, LMS, VLE • Social network • Blog • Social bookmark • Mobile app
Connect	• Call or tag other learner or teacher related to the learning topics or course • Access to online database, repositories or knowledgebase • Add new friend related to the learning topics or course	• Social network • Blog • Social bookmark • Mobile app

At the end of each session, the agent will report the percentage of learning time (P_L) to the learners, according to the frequency needed for their self-regulated learning and monitoring. For example, a learner may need to know this report at daily basis compared to his/her friend who needs it at weekly basis, depending on their goals, self-efficacy, and impact on affect and motivation needs [6]. In a long run, the report will be compiled and presented in a graphical format that learners could refer to for a long-term monitoring purpose.

TABLE IV. PARAMETERS FOR ONLINE LEARNING TIME CALCULATION

Parameter	Description	Equation
Learning time, T_L	Time spent on university learning portal or repositories (i.e. LMS, VLE, Moodle, etc.)	T_L = learning time – idle time (e)
Play time, T_P	Time spent on playing games (i.e. online and offline games)	T_P = play time – idle time (e)
Online social time, T_S	Time spent on social media, could be both for learning or playing (i.e. social network, blog, microblog, chat, etc.); can be considered as 50% for learning purposes	T_S = online social time – idle time (e)
Idle time, e	Duration of time the online usage is idle or no activities detected during the learning time, play time or online social time (i.e. no clicking, no scrolling, no typing, etc. for more than certain duration e.g. 20 minutes); can be considered as error	
Total time spent online, T_O	Learning time + Play time + Online social time	$T_O = T_L + T_P + T_S$
Percentage of learning time, P_L	Learning time + (Online social time / 2) / Total time spent online x 100	$P_L = (T_L + T_S/2) / T_O \times 100$

V. DISCUSSION

Refining the explanation in the previous section, software agents are assigned to mediate the tasks of:

- connecting to the source that the learner visits, i.e. knowledge source governed by the university, and social media tools on the Internet;
- getting the information or knowledge from the learner's online activities;
- understanding which information that is retrieved can be used to be considered as learning time (T_L), online social time (T_S), play time (T_P), idle time (e); and
- sharing the information or knowledge back to the human learner, summarising the analysis with the calculation of total online time (T_O) and percentage of learning time (P_L), for reporting purpose.

Figure 5 illustrates these tasks assignment in a nodal approach view, in which four agents are assigned with the GUSC tasks as follows:

i. **Learning Agent, A_L**: Connecting (C) to the university knowledge source (e.g. LMS, VLE and Moodle), to get (G) and understand (U) from the source, and identify the information retrieved as learning time (T_L) and idle time (e), and share (S) that knowledge to Reporting Agent, A_R.

ii. **Social Agent, A_S**: Connecting (C) to the knowledge source in the Internet and social media (e.g. social network, blog, microblog and chat), to get (G) and understand (U) from the source, and identify the information retrieved as online social time (T_S) and idle time (e), and share (S) that knowledge to both Play Agent, A_P, and Reporting Agent, A_R.

iii. **Play Agent, A_P**: Retrieving (G) the information or knowledge shared by Social Agent, A_S, analyse (U) that knowledge to identify play time (T_P), and share (S) that knowledge to Reporting Agent, A_R.

iv. **Reporting Agent, A_R**: Retrieve (G) the knowledge from Learning Agent (A_L), Social Agent (A_S) and Play Agent (A_P), analyse (U) by compiling the knowledge, and present (S) that compilation of report to the human learner (i.e. in a graphical format as suggested).

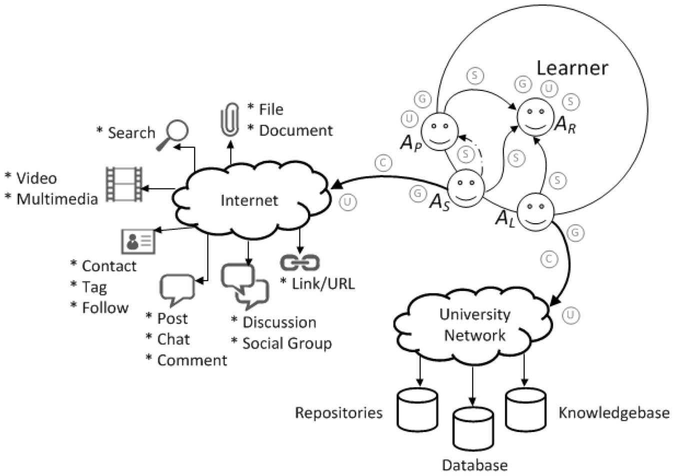

Fig. 5. Agents assigned with tasks based on source locations.

VI. Conclusion

This paper presents the conceptual model of an agent-mediate SRL, which is believed to be beneficial to learners of the current and future generation. This work needs further research in terms of its technicality and development of the multi-agent system, and further on to be tested in a real-life environment. It will be a challenging attempt but is not impossible to be implemented.

Future work should extend this conceptual model to include software agents that roam the mobile applications, since the new trend is towards mobility of the social media tools, in which the tools often reside in mobile devices. With the emerging concept of Agents of Things [28], this is deemed possible.

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